POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
College of Computer and Information Sciences

Finite Automata for Inclusive Mobility: An Assistive Navigation Tool for Users with Visual and Hearing Impairments

Automata and Language Theory Final Project Proposal

Ashley Faye J. Magcamit
Catherine Joy R. Pailden
Cristina C. Villasor
Alexandre C. Pornea

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1 Background of the Study

As per the definition from GeeksForGeeks, Finite Automata (FA) are abstract machines used to recognize patterns in input sequences, forming the basis for understanding regular languages in computer science. [3] Despite the existence of Batas Pambansa Blg. 344, inaccessibility is still an issue here in the Philippines, most facilities are not designed to accommodate People with Disabilities (PWD). [7] As a matter of fact, a documentary conducted by Rappler year (2023) found that only 5 of the 20 LRT1 stations have elevators, leveled platforms, and PWD-accessible toilets. [10] These are just some of the challenges of the disabled minority in this country regarding facilities lacking accommodations for PWDs; the actual mode of transportation is a whole other problem as well. A study by Pajarin et al. named “Assessment of Mobility of Persons with Disabilities (PWDs) in Cainta, Rizal” found that every mode of transportation vehicles failed to include PWD accessibility in their design. [8]

With this in mind, the researchers came up with the idea of creating an application that employs Finite Automata to assist users in inaccessible environments. This application aims to provide real-time guidance and alerts to users who may have visual or auditory impairments. It scans the visual from the environment to help users stay on track or find alternative routes. Aside from pathfinding, the application also offers to analyze the depth, width, type of road, heat maps, and type of facilities based on the visual input. The application also processes the audio data to give visual or tactile alerts to the user. The ability of Finite Automata to recognize patterns allows the application to process the data and give the appropriate alerts and guidance.

The researchers aim to address the challenges faced by individuals with disabilities regarding navigation and safety. By using Finite Automata, this application will be able to assist the users and guarantee them independence. This initiative aligns with the broader goal of promoting inclusivity and accessibility, particularly in a country like the Philippines where infrastructure and transportation systems often fall short of accommodating the needs of PWDs. By developing innovative solutions like this, researchers can contribute to a more equitable and accessible society for all and the advancement of academic research in the field of Computer Science by fostering innovation.

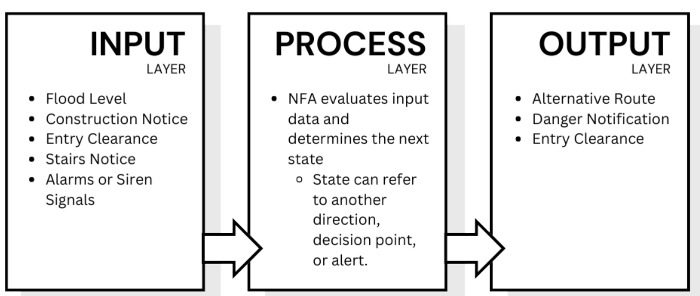

Figure 1: This figure shows the application of Nondeterministic Finite Automata (NFA) in the application through a diagram. The Input Layer collects the data from the user’s environment through input devices. These visual and audio data include: flood level, construction notice, entry clearance, stairs notice, alarms, or siren signals. The data will then proceed to the Process Layer where NFA is applied. NFA evaluates the data and decides on which state to go next. Each state corresponds to a different decision or direction which allows the application to issue alerts and alternative routes for the user.

2 Review of Related Literature

2.1 Shortest Path Problem

Several studies use the concept of shortest path problem with the FSA concept. The work of Septiani et al. [11] used the concept of finite state automaton (FSA) to find the shortest path for Margonda and Univeritas Indraprasta, where they successfully solved the single-source path problem using the FSA concept. They, however, did not show the complexity of their proposed algorithm to determine if it is efficient. Another, Huguet et al. [1] proposes an algorithm for finding the shortest path for different path problems using FSA that results in polynomial complexity and maintains a high-speed process despite many nodes or points. Its limitation of more nodes may equate to a complex computation, leading to a slow process.

2.2 Wayfinder Model

In the study of [2] and [9], Most people with visual impairments and blindness face challenges because of the different aspects of human behavior, such as sensory, behavioral, psychological, and analytical. Furthermore, as the number of studies that focus on pathfinding for visually impaired and deaf people is large, most of the studies were conducted in a controlled environment such as laboratories and hospitals, which do not include technologies that are based on real-world scenarios. [5] also stated that most of the researchers tend to use participants who are blindfolded instead of people with vision impairments. According to [4], the researchers explored the pathfinding process by analyzing and identifying the type of environment based on the preference of the visually impaired users. Most of the studies provided verbal guidance for wayfinding and described their surroundings, such as the compositions of the buildings, roads, and overpasses to reach their destinations efficiently.

2.3 Smart Assistive System

In existing studies of the developed smart assistive system for the visually impaired, the noticeable gap is the lack of suggested alternative routes for the visually impaired individual when navigating their environment. This is crucial especially for the country of Philippines which is not carefully constructed for PWDs accessibility. In the study of [6] they've developed an effective system that aims to assist impaired people with object detection and scene classification. They utilized their methodology using Raspberry-Pi 4B, Camera, Ultrasonic Sensor and Arduino, mounted on the stick of the impaired individual. To analyze the scene, they take pictures of the scene and afterwards process using techniques to detect objects by the help of Viola Jones and TensorFlow Object Detection algorithm. In measuring the distance of the impaired individual and object, they employed an ultrasonic sensor mounted on a servomotor to measure the distance of the individual to the obstacle. [2] also utilized the ultrasonic sensor in their proposed Smart Assistive aid wherein they aim to improve the accessibility of visually impaired individuals by safely navigation through object detection and text recognition using the Raspberry-Pi 4B v2 camera module.

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